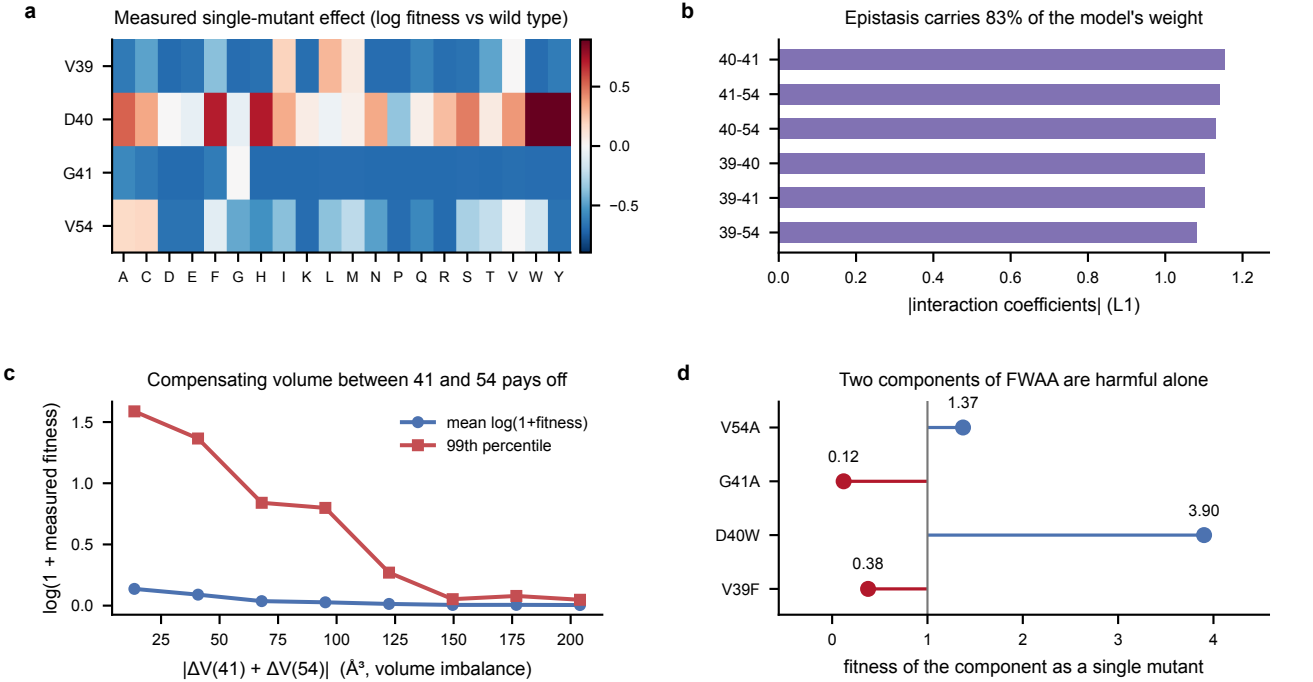

a
Measured single-mutant effect (log fitness vs wild type)
V39
D40
G41
V54
A C D E F G H I K L M N P Q R S T V W Y
0.5
0.0
−0.5
b
Epistasis carries 83% of the model's weight
40-41
41-54
40-54
39-40
39-41
39-54
0.0 0.2 0.4 0.6 0.8 1.0 1.2
|interaction coefficients| (L1)
c
Compensating volume between 41 and 54 pays off
mean log(1+fitness)
99th percentile
log(1 + measured fitness)
1.5
1.0
0.5
0.0
25 50 75 100 125 150 175 200
|ΔV(41) + ΔV(54)| (Å³, volume imbalance)
d
Two components of FWAA are harmful alone
V54A
G41A
D40W
V39F
1.37
0.12
3.90
0.38
0 1 2 3 4
fitness of the component as a single mutant